



Introduction to AI



Outline

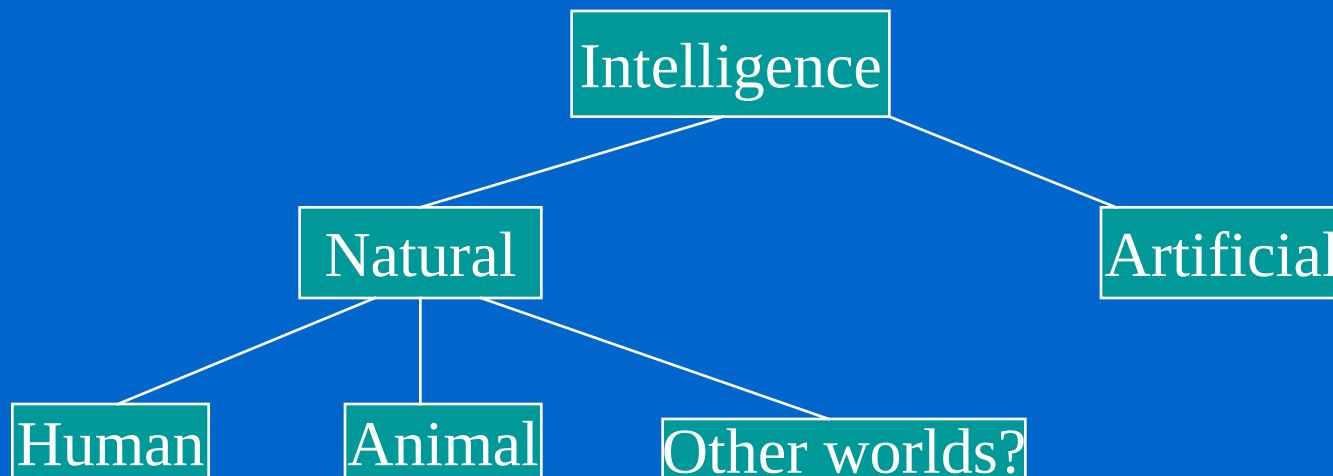
- What is Artificial Intelligence?
- Some Achievements
- AI's Genesis
- Overview of Problems and Techniques
- The major sub-areas of AI
- Summary
- Exercise

• • • Introduction to Artificial Intelligence

“There are three great events in history. The creation of the universe. Two, the appearance of life. The third one, which I think is equal in importance, is the appearance of artificial intelligence” Edward Fredkin

? What is intelligence?

? Is it possible for machine to be intelligent?



Introduction

Big questions

- Can machines think? [../AI/computing_machinery_and_intelligence - a_m_turing, 1950.htm](http://AI/computing_machinery_and_intelligence_-_a_m_turing_1950.htm)
- And if so, how?
- And if not, why not?
- And what does this say about human beings?
- And what does this say about the mind?
- Is it possible for machines to be intelligent?

What is Intelligence?

- Webster's Dictionary

The faculty of acquiring and applying knowledge

- More scientific

Intelligence is a measure of the success/performance of an entity in achieving its ***objectives/goals*** by interaction with its ***environment***.

- The *level of intelligence* should be judged by considering the ***difficulty of the goals***, and the **success** in achieving them
- Intelligent quotient (amount, percentage) is measured by tests, involving a variety of skills and knowledge
 - Do all antonyms of the following words start with 'c'?
Open, courage, etc.

What is Intelligence?

The definition mentioned in the previous slide has some important consequences:

- Intelligent behaviour can only be observed in the presence of an *environment*
- To measure intelligence there must be **goals** and a **scale** to determine how well these are achieved.
- The ability to “express” intelligence depends on the “richness” of interaction with the environment, and on the subtlety (delicacy) of the goals, as well as internal mechanisms
- This definition clearly allows the possibility of *intelligent machine*.

Aspects/Characteristics of Intelligence

- An intelligent entity should have the ability to:
 - interact with its environment;
 - learn from information and experience;
 - make rational decisions;
 - deduce new facts from given ones
 - make sensible deductions when insufficient facts are available

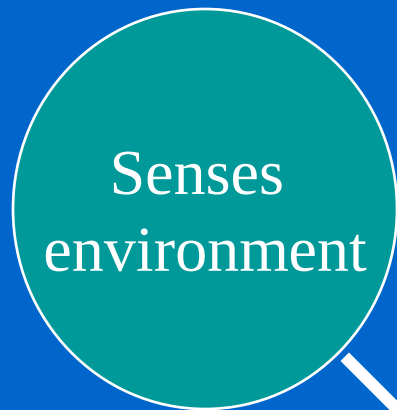
This implies some form of getting input (i.e. senses), a way to produce output and an ability to process the inputs to give the output some relevance

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See

Hear
Touch
Taste
Smell

Inputs

Internal Processes

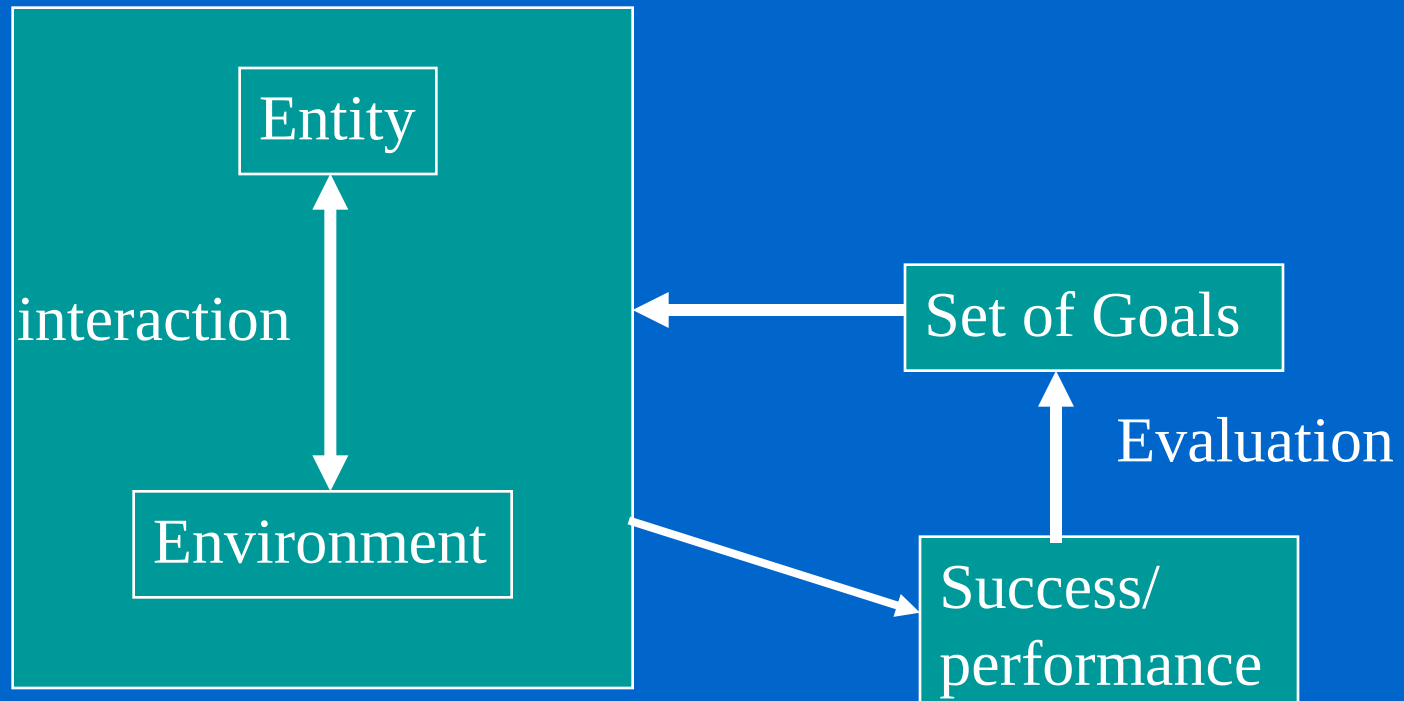


outputs

**An
Intelligent
Entity**

Scale of Intelligence

- To be able to measure intelligence, an environment and set of goals must clearly be defined
- For example if intelligence is only measured on an entities ability to avoid obstacles then mice would prove to be more intelligent than a person who is blindfolded



Components of Intelligence

Signs of Intelligence

- Learn or understand from experience
- Make sense out of ambiguous or contradictory messages
- Respond quickly and successfully to new situations
- Use reasoning to solve problems
- Understand and infer in ordinary, rational ways
- Apply knowledge to manipulate the environment
- Think and reason
- Recognize the relative importance of different elements in a situation

Intelligent Machines: The Behaviorist View

- Is it possible for machines to be intelligent?
 - Machines can exhibit intelligent behaviour but how intelligent are they really and is this the same as “human intelligence”?
 - Many scientists believe that only things that can be directly observed are “scientific” (behaviourism)
 - Therefore, if a machine behaves “as if it were intelligent”, it is meaningless to argue that this is an illusion
 - This view can be summarised as: “ It it walks like a duck, quacks like a duck and looks like a duck - it is a duck”
 - Turing was of this opinion and proposed the “Turing Test”

What is artificial intelligence?

- There are no clear consensus on the definition of AI

Q. *What is artificial intelligence?*

A. It is the science and engineering of making intelligent machines, **especially intelligent computer programs**. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.

Q. *Yes, but what is intelligence?*

A. Intelligence is the computational part of the **ability to achieve goals in the world**. Varying kinds and degrees of intelligence occur in people, many animals and some machines.

Other possible AI definitions

- AI is a collection of hard problems which can be solved by humans and other living things, but for which we don't have good algorithms for solving.
 - e. g., understanding spoken natural language, medical diagnosis, learning, self-adaptation, reasoning, chess playing, proving math theories, etc.
- Definition from R & N book: a program that

Systems that act like human

Systems that think like human

Human-centred approach must be an empirical science, involving hypothesis and experimental confirmation

Systems that act rationally

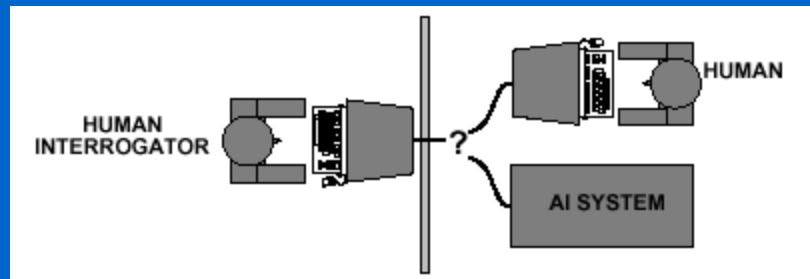
Systems that think rationally

Rationalist approach involves a Combination of mathematics and engineering

Acting Humanly: The Turing Test

Turing (1950s): Intelligent behavior as the ability to achieve human-level performance in all cognitive tasks.

Interrogating the system via a teletype without physical interaction.



Systems need to possess : Knowledge, reasoning, language understanding and learning.

The Turing Machine

- A universal theoretical model of computation
- Proposed by Alan Turing (1950) designed to provide a satisfactory **operational definition** of intelligence
- Turing defined intelligent behaviour as the ability to achieve human level performance in all cognitive tasks
- The approach deliberately avoids direct physical interaction between the interrogator and the computer because physical simulation of a person is unnecessary for intelligence

The Turing Test (Acting humanly)

- A judge has to find out which of two hidden agents is a human and which is a machine.
- To pass total Turing test computer need
 - computer vision to perceive objects
 - robotics to move the objects

Thinking Humanly: Cognitive Process

1960s “cognitive revolution”: information-processing psychology replaced behaviorism.

Understanding how humans think requires scientific theories of internal activities of the brain

1. What level of abstraction? ‘Knowledge’ or ‘circuits’?
2. How to validate? Requires
 1. Predicting and testing behavior of human subjects
 2. Identification of neurological data.

Cognitive science and cognitive neuroscience are distinct from AI but share one principal direction.

Rules of thought

- Logic: Aristotle developed the first formal approach to reasoning in his syllogism.
 - All Greeks are mortal
 - Socrates is a Greek
 -
 - Socrates is mortal

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Acting Rationally

Acting rationally = to achieve one's goals, given one's beliefs.

Difference between acting and thinking rationally: correct inference is not *all* of rationality.

Need ability to represent knowledge and reason with it.

Reach good decisions in a wide variety of situations.

AI: study and construction of rational agents

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Rational Agents

An agent is an entity that perceives and acts.

For any given class of environments and tasks, we seek the agent (or class of agent) with the best performance.

Computational limitations make perfect rationality unachievable. Design best program of given machine resources.

This class is about construction rational agents.....

AI Objectives

- The objective of AI is to **understand and model cognition, intelligence, and intelligent behaviour.**
- Write the names of three skills.

Think what is required to do this?

- Counting
- Calculating
- Doing Mathematics
- Painting a picture
- Recognising a face of a friend
- Understanding a story or a fairy tale.
- Reading newspapers
- Making decisions
- Finding the shortest tour to visit a number of places
- Writing a program
- Chatting at a party to other guests

Some of AI's achievements

- In 1997, the chess program Deep Blue of IBM defeated the chess world champion Garry Kasparov in a match under regular tournament conditions
- Expert systems are routinely employed in a variety of industries
- Machines manage to understand (recognise) spoken language
- *Write ten others (assignment)*

AI's Genesis

- From philosophy - theories of reasoning and learning have emerged
- From Mathematics - formal theories of logic, probability, decision-making and computation
- From Psychology - tools to investigate human mind
- From Linguistics - theory of the structure and meaning of language
- From Computer Science - tools with which to make AI a reality

AI's Genesis

- For long have people wondered about how the human mind works, how to judge properly, how to act wisely, etc.
- The ancient Greek philosophers were already quite advanced in their analysis of the ***foundations of our human intelligence***
- They were concerned with the ***substance and composition of the world*** as well as with the way how we humans perceive and understands the world around us and how to love a good life
- Plato and Aristotle (about 400 BC) laid the foundations of ***two opposing views*** of how humans come to their concepts to understand the world around us

Rationalism

- Rationalism (Plato) is the idea that our concepts come from the ratio (relationship) , i.e. from our mind and *we just apply* our natural concepts to our sense experience.
- Plato spoke of remembering the “natural/innate” concepts of the world system when we perceive objects as belonging to a certain kind (Object Model).



Empricism

- Empricism (Aristotle) is the idea that our concepts to classify things are derived from our senses experiences i.e. we shape our concepts by perceiving the world system around us and gradually learn to cope with the world



The ‘critical turn’

- The German philosopher Kant (18th century) presented a foundational analysis in his “Critique of Pure Reason”, which essentially rejects both, Rationalism and Empiricism, and shows that a synthesis of both is necessary to account for the process underlying human intelligence

AI's Genesis

- A logical reasoning calculus was conceived and a calculating machine built by Leibniz in the 17th century
- Leibniz wanted to resolve intellectual arguments by calculation
- 1847: Boole developed “Boolean logic”
- 1897: Frege developed today’s Predicate Logic in his Begriffsschrift.
- 1931: Godel proved his incompleteness theorem

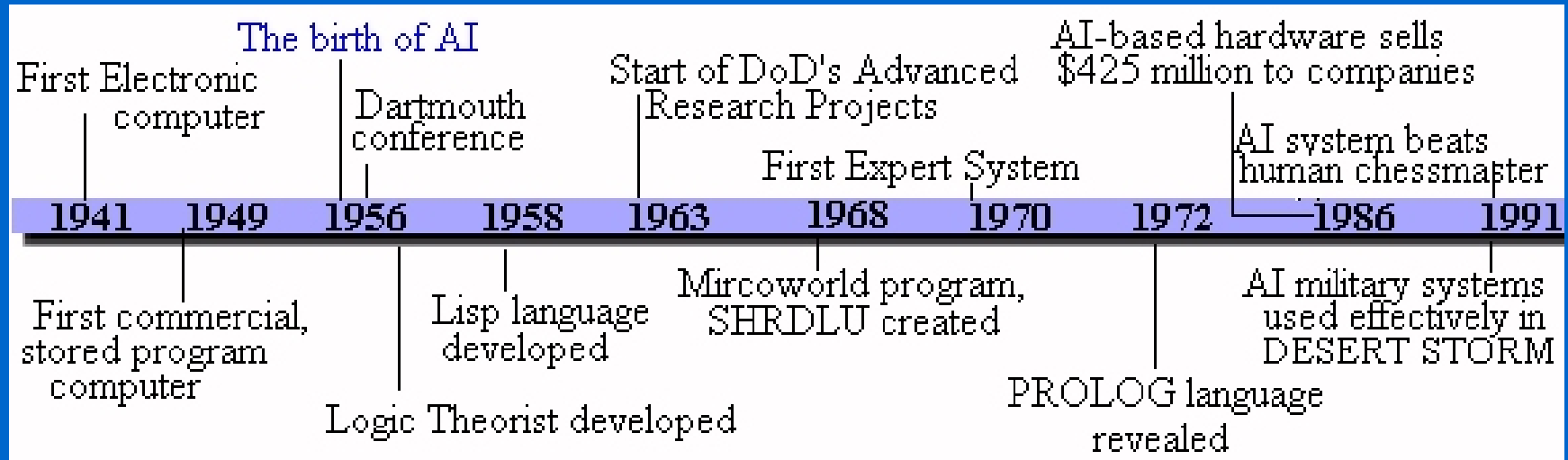
History of AI

- AI has roots in a number of scientific disciplines
 - computer science and engineering (hardware and software)
 - philosophy (rules of reasoning)
 - mathematics (logic, algorithms, optimization)
 - cognitive science and psychology (modeling high level human/animal thinking)
 - neural science (model low level human/animal brain activity)
 - Linguistics, economics, etc.
- The birth of AI (1943 – 1956)
 - Pitts and McCulloch (1943): simplified mathematical model of neurons (resting/firing states) can realize all propositional logic primitives (can compute all Turing computable functions)
 - Allen Turing: Turing machine and Turing test (1950)
 - Claude Shannon: information theory; possibility of chess playing computers
 - Tracing back to Boole, Aristotle, Euclid (logics, syllogisms)

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- Early enthusiasm (1952 – 1969)
 - 1956 Dartmouth conference
 - John McCarthy (Lisp);
 - Marvin Minsky (first neural network machine);
 - Alan Newell and Herbert Simon (GPS);
 - Emphasize on intelligent general problem solving
 - GSP (means-ends analysis);
 - Lisp (AI programming language);
 - Resolution by John Robinson (basis for automatic theorem proving);
 - heuristic search (A^* , AO^* , game tree search)
- Emphasis on knowledge (1966 – 1974)
 - Domain specific knowledge is the key to overcome existing difficulties
 - Knowledge representation (KR) paradigms

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- Knowledge-based systems (1969 – 1979)
 - DENDRAL: the first knowledge intensive system (determining 3D structures of complex chemical compounds)
 - MYCIN: first rule-based expert system (containing 450 rules for diagnosing blood infectious diseases)
EMYCIN: an ES shell
 - PROSPECTOR: first knowledge-based system that made significant profit (geological ES for mineral deposits)
- AI became an industry (1980 – 1989)
 - wide applications in various domains
 - commercially available tools
- Current trends (1990 – present)
 - more realistic goals
 - more practical (application oriented)
 - distributed AI and intelligent software agents
 - resurgence of neural networks and emergence of genetic algorithms

History



Overview: Problems and Techniques

- Lack of early AI researcher's understanding of the problem tackled
- Lack of knowledge in the machine
- Difficulty in both formalizing everyday knowledge as well as expert knowledge

State of the Art

Which of the following can be done at present?

1. Play a decent game of bridge.
2. Understand human speech.
3. Discover and prove a new mathematical theorem.
4. Write a funny story.
5. Translate spoken English into spoken Swedish in real time.
6. Perform a complex surgical operation.
7. Drive along a curving mountain road.

Summary

- The objective of AI is to understand and model cognition, intelligence and intelligent behaviour.
- This is not a homogeneous objective. Different researchers have different interests which overlap and have different foci in the field of AI
- Important aspects of this overall objective have been studied in Computing, Cybernetics, Linguistics, Mathematics, Philosophy and Psychology

Summary

- AI has considerably matured over the past decades and is now becoming increasingly important from a commercial point of view
- So far, AI has had a number of impressive success stories
- Some techniques went more or less unacknowledged into main stream Computer Science, such as Object Oriented Programming
- But the real impact lies still ahead of us.

Assignment

- Write ten definitions of AI
- Write ten achievements of AI
- Solve the following exercises from R&N
 - 1.1,1.2,1.3,1.5, 1.6, 1.7, 1.9 and 1.10



Intelligent Agents



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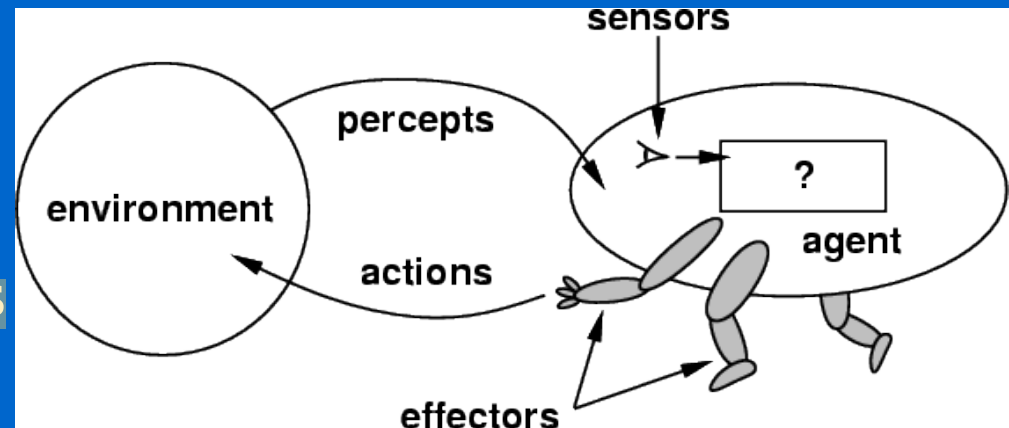
University of Chittagong

Lecture Outlines

- Define Intelligent Agents

Intelligent Agents

- Definition: An Intelligent Agent/entity that perceives its environment via sensors and acts rationally upon that environment with its effectors.
- Example: human, robotic, software agents
- Hence, an agent gets percepts one at a time, and maps this percept into sequence to actions.
- Properties
 - Autonomous
 - Interacts with other agents plus the environment
 - Reactive to the environment
 - Pro-active (goal-directed)



What do you mean, sensors/percepts and effectors/actions?

- Humans
 - Sensors: Eyes (vision/act of seeing), ears (hearing), skin (touch), tongue (gustation/sense of taste /salt), nose (olfaction/the sense of smell), neuromuscular system (Proprioception is an automatic sensitivity mechanism in the body that sends messages through the central nervous system (CNS). The CNS then relays information to rest of the body about how to react and with what amount of tension)
 - Effectors: limbs, digits, eyes, tongue, mouth ...
 - Actions: lift a finger, turn left, walk, run, carry an object, ...
- The Point: percepts and actions need to be carefully defined, possibly at different levels of abstraction

How Agents Should Act

- A rational agent is one that does the right thing.
- That is, the rational agent will act such that it will be successful.
- A **performance measure** determines how successful an agent is.

Performance Measures

- Subjective vs objective
- An outside measure may be more objective
- To create a sensible performance measure is often rather difficult

Examples of Performance Measures

- Vacuum cleaner: how much dirt removed
- Call centre: how many calls handled?
- Teaching: how many students passed?

Rationality versus Omniscience

- An omniscience agent knows everything, including the outcome of the actions
- A rational agent works with “reasonable expectations” based on what has been perceived
- Example
 - Seeing an old friend across the street

What is rational is determined by

- The **performance measure** that defines **degree of success**
- The **perceptions of the agents** so far (the percept story)/percept sequence
- The **agent's knowledge** on the environment
- The actions the agent can perform

An Ideal Rational Agent

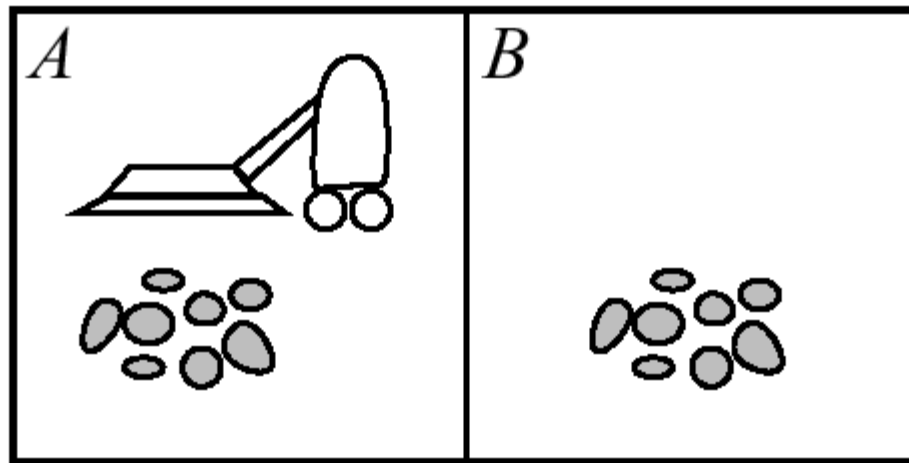
- An ideal **rational agent** should, for each possible percept sequence, do whatever actions that will maximize its performance measure based on
 - (1) the percept sequence, and
 - (2) its built-in and acquired knowledge.
- This includes information gathering as a rational activity
- Examples
 - Crossing the road
 - Clock

Rationality: maximizes expected performance. Doesn't mean success every time.

Rationality $\neq$ perfection Rationality $\neq$ Success always.

- Types of performance measures: payoffs, false alarm and false dismissal rates, speed, resources required, effect on environment, etc.

Simple example: Vacuum-cleaner world



Percepts: room and its cleanness, e.g. [B, clean]

Actions: Go Left, Go Right, Vacuum, Do nothing.

In this simple world, the function f (the agent function) is simple. We can tabulate all function values in many ways.

For example:

Percept Sequence

[A, clean]

[A, dirty]

[B, clean]

[B, dirty]

[A, clean], [A, clean]

[A, clean], [A, dirty]

Action

(Go Right)

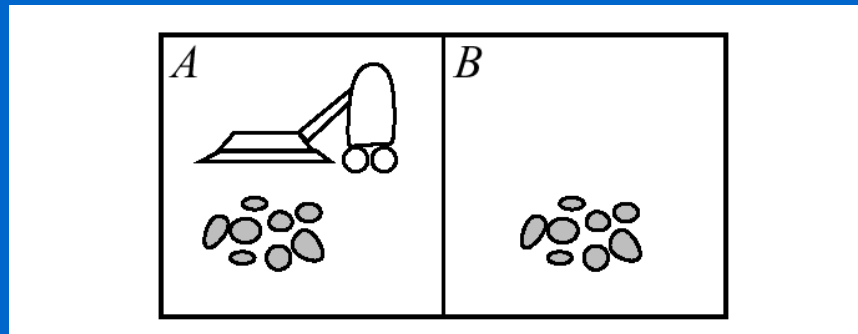
(Vacuum)

(Go Left)

(Vacuum)

(Go Right)

(Vacuum)

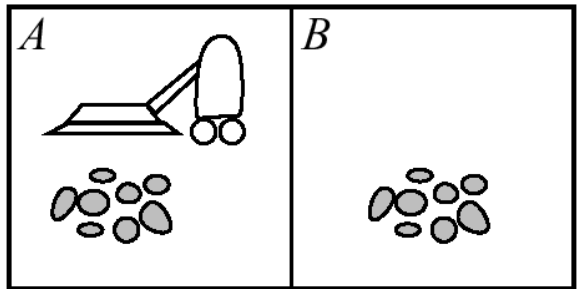


Our task: fill out the table in a correct way so that the vacuum cleaner looks intelligent.

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Criterion for success of an agent's behavior. Keeping scores

- One point per ounce of dirt cleaned.
- One point per square cleaned up in time T.
- One point per clean square per time step, minus one per move.



What you ask for is what you get.... If you really know what ask for.

• : Ideal Mapping from Percept Sequence to Actions

- Square root function example
- lookup table

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Mapping Percept Sequences to Actions - Lookup Table

Percept x	Action z
1.0	1.000
1.1	1.048
1.2	1.095
1.3	1.140
1.4	1.183
1.5	1.224
1.6	1.264
1.7	1.303
1.8	1.341
1.9	1.378

Mapping Percept Sequences to Actions

Percept x	Action z	
1.0	1.000	
1.1	1.048	Function
1.2	1.095	function SQRT(x)
1.3	1.140	$z := 1.0$
1.4	1.183	repeat until $ z * z - x < 0.001$
1.5	1.224	$z := z - (z * z - x) / (2z)$
1.6	1.264	end
1.7	1.303	return z
1.8	1.341	
1.9	1.378	

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Autonomy

- A system is autonomous to the extent that its own behavior is determined by its own **experience and knowledge**.
- Therefore, a system is not autonomous if it is guided by its designer according to a priori decisions.
- To survive agents must have:
 - Enough built- in knowledge to survive.
 - Ability to learn.

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We have discussed agents by describing their **Behavior (external)** – the **action** that is performed after any **Given sequence of precepts**.

∴ Structure of Intelligent Agents

- The job of AI is to design the agent program
 - the program is a function that implement the agent mapping from percepts to actions
 - the program will run on some sort of computing device, which we will call the architecture
 - the architecture might be a plain computer, it might also include software that provide a degree of insulation between the raw computer and agent program

AGENT = ARCHITECTURE + PROGRAM

architecture make available percepts received from sensor to the agent program, it runs the program and feeds the program's actions to the effectors

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Building a rational agent..... first step.

Specify the *task environment* : performance measure, the environment, agent's actuators and sensors.

PEAS

Building a rational robot professor.....

Performance measure/goals : Student evaluation(?) Average class grade (?) Average final exam grade(?)

Environment: hostile students (?) friendly students (?) well-equipped classroom (?), outdoor classroom(?), no TA(?)

Actuators: speech mechanism(?), projectors (?)

Sensors: cameras(?), voice recorders(?)

Examples of Agent Types and their Descriptions

Agent Type	Percepts	Actions	Goals	Environment
Medical diagnosis system	Symptoms, findings, patient's answers	Questions, tests, treatments	Healthy patient, minimize costs	Patient, hospital
Satellite image analysis system	Pixels of varying intensity, color	Print a categorization of scene	Correct categorization	Images from orbiting satellite
Part-picking robot	Pixels of varying intensity	Pick up parts and sort into bins	Place parts in correct bins	Conveyor belt with parts
Refinery controller	Temperature, pressure readings	Open, close valves; adjust temperature	Maximize purity, yield, safety	Refinery
Interactive English tutor	Typed words	Print exercises, suggestions, corrections	Maximize student's score on test	Set of students

AGENT PROGRAM

Function SKELETON-AGENT (percept) returns action //single percept
Static: memory // the agent's memory of the world

```
memory = UPDATE-MEMORY (memory, percept)
action = CHOOSE-BEST-ACTION (memory)
memory = UPDATE-MEMORY (memory, action) //action is also
stored
return action
```

Performance measure is not the part of this program as it is applied
Externally to judge the behaviour of the agent.

Some Agent Types

5 agent Types

- **Table-driven agents**
 - use a percept sequence/ action table in memory to find the next action. They are implemented by a (large) lookup table.
- **Simple reflex agents**
 - are based on condition-action rules and implemented with an appropriate production (rule-based) system. They are stateless devices which do not have memory of past world states.
- **Agents with memory**
 - have internal state which is used to keep track of past states of the world.
- **Agents with goals**
 - are agents which in addition to state information have a kind of goal information which describes desirable situations. Agents of this kind take future events into consideration.
- **Utility-based agents**
 - base their decision on classic axiomatic utility-theory in order to act rationally.

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A more specific example: Automated taxi driving system

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- **Percepts:** Video, sonar, speedometer, odometer, engine sensors, keyboard input, microphone, GPS, ...
 - **Actions:** Steer, accelerate, brake, horn, speak/display, ...
 - **Goals:** Maintain safety, reach destination, maximize profits (fuel, tire wear), obey laws, provide passenger comfort, ...
 - **Environment:** urban streets, freeways, traffic, pedestrians, weather, customers, ...
 - **Different aspects of driving may require different types of agent programs!**

Table Driven Agent

Function TABLE-DRIVEN-AGENT (percept) returns action

Static: percepts // a sequence, initially empty

table // a table, indexed by percept sequences, initially fully specified

append *percept* to the end of *percepts*

action = LOOKUP (percepts, table)

return action

Limitation

- the table needed for something as simple as an agent that can play chess would be about 35^{100} entries
- it would take quite a long time for the designer to build the table
- no autonomy

Simple Reflex Agent

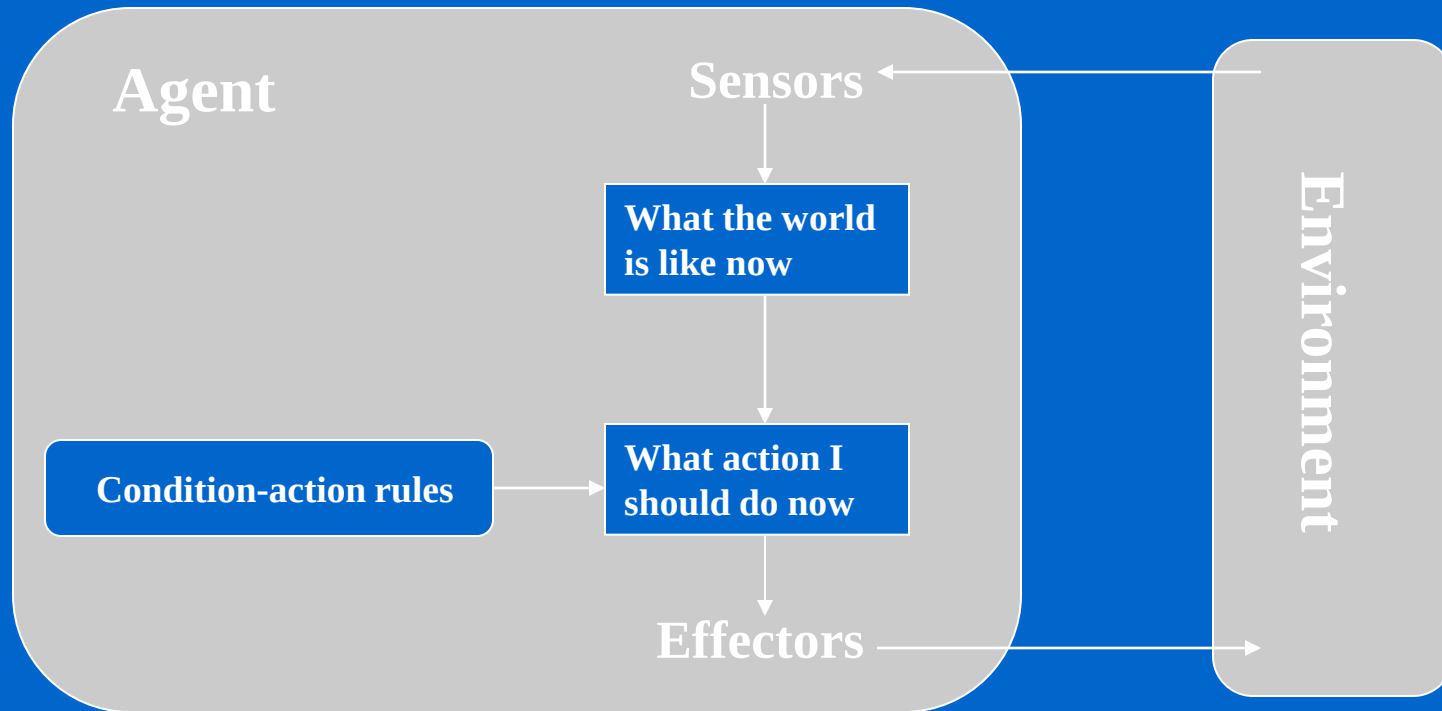
- Use *condition-action* rules to summarize portions of the table

**If car-in-front-is braking (condition)
Then initiate braking (action - rule)**

Function SIMPLE-REFLEX-AGENT (percept) returns action
Static: rules // a set of condition-action rules

```
state = INTERPRET-INPUT (percept) //generate an abstracted
                                   description of the current state from the percept
rule = RULE-MATCH (state, rules) /
action = RULE-ACTION [Rule]
return action
```

A Simple Reflex Agent: Schema



Act on current percept, ignoring percept history

Reflex Agent with Internal State

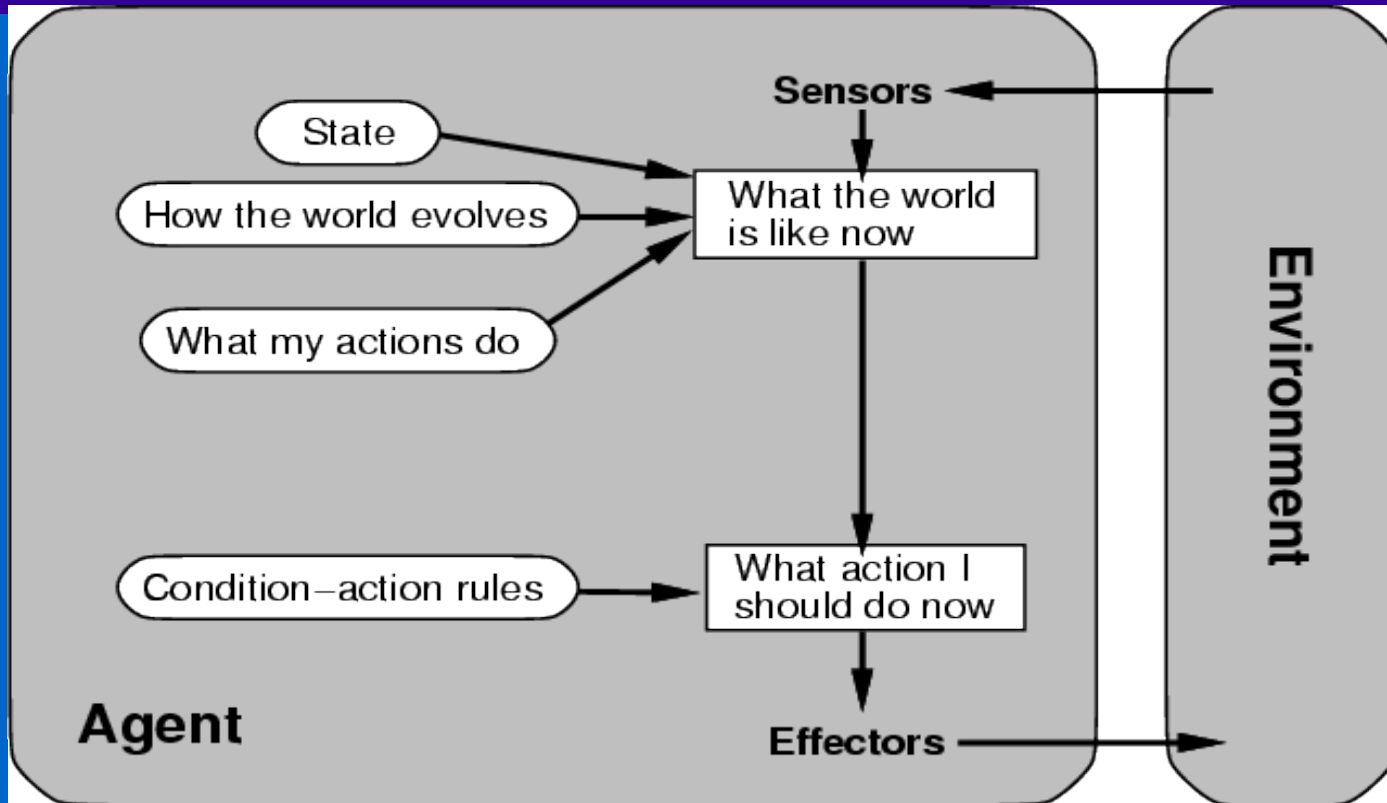
- Encode "internal state" of the world to remember the past as contained in earlier percepts
- Needed because sensors do not usually give the entire state of the world at each input, so perception of the environment is captured over time. "State" used to encode different "world states" that generate the same immediate percept.
- Requires ability to represent change in the world; one possibility is to represent just the latest state, but then can't reason about hypothetical courses of action

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Function REFLEX-AGENT (percept) returns action
Static: state // a description of the current world state
rules // a set of condition-action rules

```
state = UPDATE-STATE (state, percept)
rule = RULE-MATCH (state, rules)
action = RULE-ACTION (rule)
state = UPDATE-STATE (state, action)
return action
```

Agents that Keep Track of the World



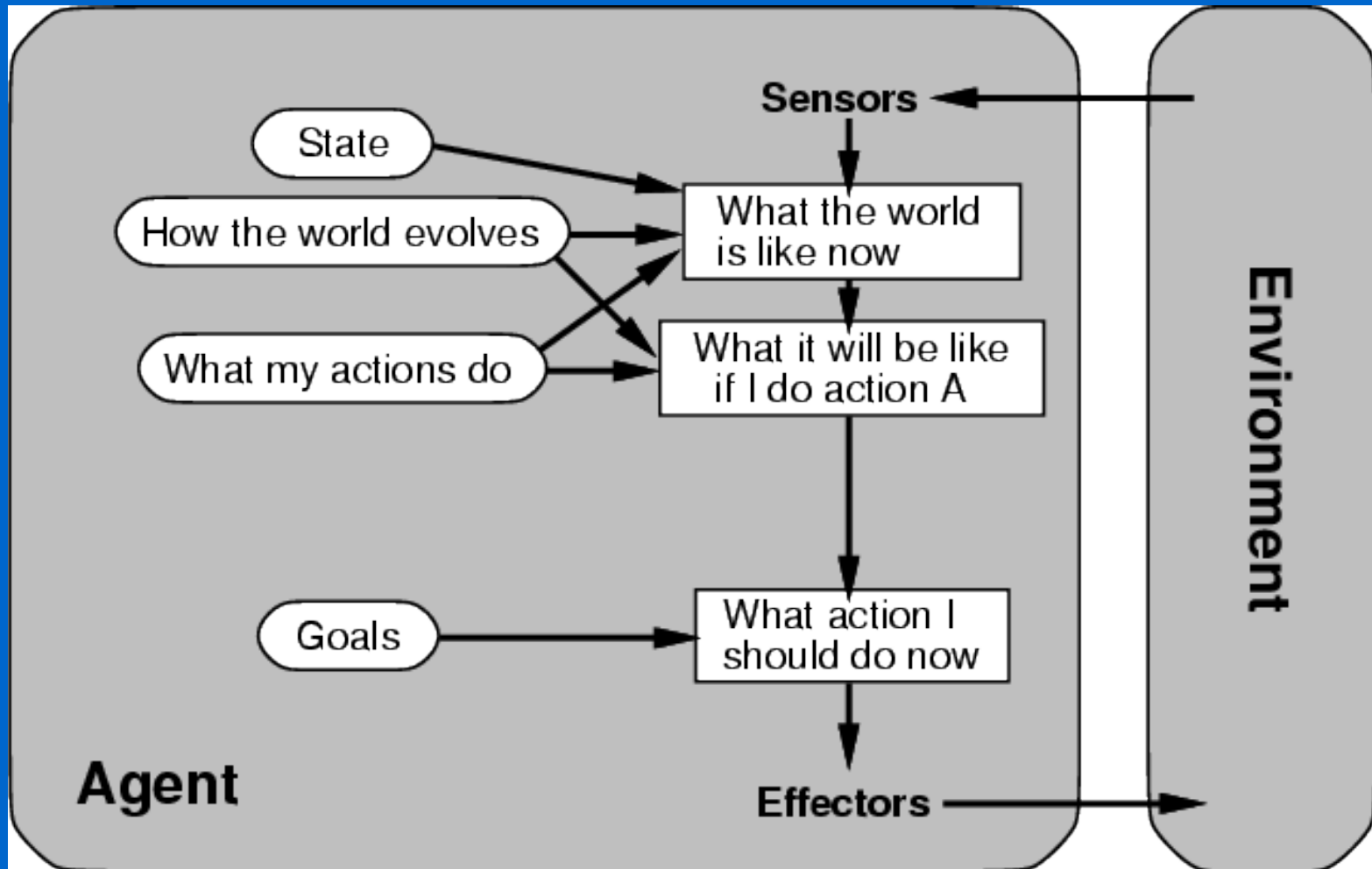
Use internal states (or models) to deal with the world that is only partially observable.

Goal- Based Agent

- Choose actions so as to achieve a (given or computed) goal.
- A goal is a description of a desirable situation
- Keeping track of the current state is often not enough -- need to add goals to decide which situations are good
- Deliberative instead of reactive
- May have to consider long sequences of possible actions before deciding if goal is achieved -- involves consideration of the future, “*what will happen if I do...?*”

Agents with Explicit Goals

Knowing the world and respond appropriately is not the whole story
Anticipating future ... involving planning and search.



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Goal-Based Agent

function AGENT-WITH-EXPLICIT_GOAL(*percept*, *goal*) **returns**
action

static: *state* // a description of the current world state

rules // a set of condition-action rules

state = UPDATE_STATE(*state*, *percept*)

rule = RULE-MATCH(*state*, *rules*, *goal*)

action = RULE-ACTION(*rule*, *goal*)

state = UPDATE-STATE(*state*, *action*)

return *action*

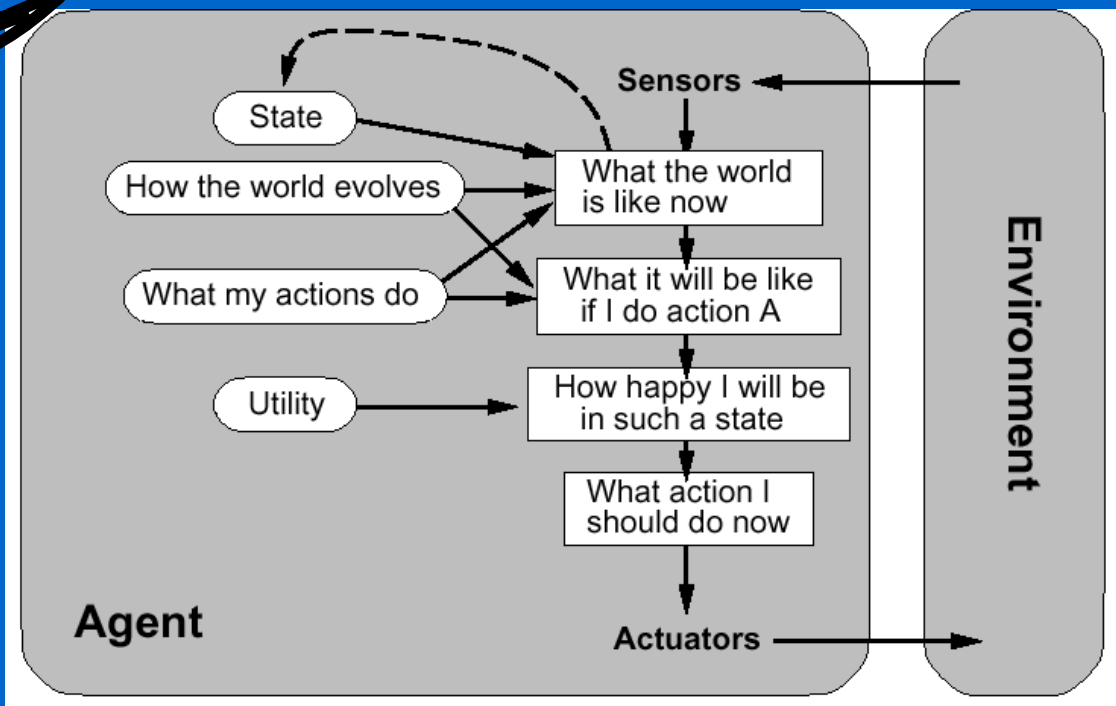
Utility- Based Agent

- When there are multiple possible alternatives, how to decide which one is best?
- A goal specifies a crude distinction between a happy and unhappy state, but often need a more general performance measure that describes "degree of happiness"
- Utility function **U**: **States** --> **Reals** indicating a measure of success or happiness when at a given state
- Allows decisions comparing choice between conflicting goals, and choice between likelihood of success and importance of goal (if achievement is uncertain)

Utility-based agents

Utility (cost) function maps a state onto a real number.

Improve upon the goal-based agents by having high-quality behavior in most environment.



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Utility-Based Agent

function UTILITY-BASED-AGENT(*percept*, *utility-function*) **returns**
action

static: *state* // a description of the current world state

rules // a set of condition-action rules

state = UPDATE_STATE(*state*, *percept*)

action = UTILITY-MAXIMISER(*state*, *rules*, *utility-function*)

state = UPDATE-STATE(*state*, *action*)

return *action*

Properties of Environments

- **Accessible/ Inaccessible.**

- If an agent's sensors give it access to the complete state of the environment needed to choose an action, the environment is accessible.
- Such environments are convenient, since the agent is freed from the task of keeping track of the changes in the environment.

- **Deterministic/ Non-deterministic.**

- An environment is deterministic if the next state of the environment is completely determined by the current state of the environment and the action of the agent.
- In an accessible and deterministic environment the agent need not deal with uncertainty.

- **Episodic/ Nonepisodic.**

- An episodic environment means that subsequent episodes do not depend on what actions occurred in previous episodes.
- Such environments do not require the agent to plan ahead.

Properties of Environments

- **Static/ Dynamic.**

- A static environment does not change while the agent is thinking.
- In a static environment the agent need not worry about the passage of time while he is thinking, nor does he have to observe the world while he is thinking.
- In static environments the time it takes to compute a good strategy does not matter.

- **Discrete/ Continuous.**

- If the number of distinct percepts and actions is limited the environment is discrete, otherwise it is continuous.

- **With/ Without rational adversaries.**

- If an environment does not contain other rationally thinking, adversary (opponent) agents, the agent need not worry about strategic, game theoretic aspects of the environment
- Most engineering environments are without rational adversaries, whereas most social and economic systems get their complexity from the interactions of (more or less) rational agents.

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Characteristics of environments

	Accessible	Deterministic	Episodic	Static	Discrete
Solitaire					
Backgammon					
Taxi driving					
Internet shopping					
Medical diagnosis					

Characteristics of environments

	Accessible	Deterministic	Episodic	Static	Discrete
Solitaire	No	Yes	Yes	Yes	Yes
Backgammon					
Taxi driving					
Internet shopping					
Medical diagnosis					

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Medical diagnosis					

Characteristics of environments

	Accessible	Deterministic	Episodic	Static	Discrete
Solitaire	No	Yes	Yes	Yes	Yes
Backgammon	Yes	No	No	Yes	Yes
Taxi driving	No	No	No	No	No
Internet shopping	No	No	No	No	No
Medical diagnosis	No	No	No	No	No

→ Lots of real-world domains fall into the hardest case!

Environment	Accessible	Deterministic	Episodic	Static	Discrete
Chess with a clock	Yes	Yes	No	Semi	Yes
Chess without a clock	Yes	Yes	No	Yes	Yes
Poker	No	No	No	Yes	Yes
Backgammon	Yes	No	No	Yes	Yes
Taxi driving	No	No	No	No	No
Medical diagnosis system	No	No	No	No	No
Image-analysis system	Yes	Yes	Yes	Semi	No
Part-picking robot	No	No	Yes	No	No
Refinery controller	No	No	No	No	No
Interactive English tutor	No	No	No	No	Yes

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Environment Programs

- Read from R&N (pages 47 – 49)

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Summary

- An **agent** perceives and acts in an environment, has an architecture and is implemented by an agent program.
- An **ideal agent** always chooses the action which maximizes its expected performance, given percept sequence received so far.
- An **autonomous agent** uses its own experience rather than built-in knowledge of the environment by the designer.
- An **agent program** maps from percept to action & updates its internal state.
 - **Reflex agents** respond immediately to percepts.
 - **Goal-based agents** act in order to achieve their goal(s).
 - **Utility-based** agents maximize their own utility function.
- **Representing knowledge** is important for successful agent design.
- Some **environments** are more difficult for agents than others. The most challenging environments are inaccessible, non-deterministic, non-episodic, dynamic, and continuous.

Assignment

- What is the difference between a performance measure and a utility function?
- For each of the environments in slide , determine what type of agent architecture is most appropriate (table lookup, simple reflex, goal-based or utility-based)
- Choose a domain that you are familiar with, and write a PAGE description of an agent for the environment. Characterize the environment as being accessible, deterministic, episodic, static and continuous or not. What agent architecture is best for this domain?
- Exercise 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 2.10, 2.11